

ServiceNow Project Documentation

Laptop Request Management – Business Rules & Security

16. Business Rule – One Laptop Request Per Year

Purpose

To ensure that an employee can submit only one laptop request per calendar year, a Business Rule was created at the backend level.

The Business Rule checks whether the requester has already submitted a laptop request during the current calendar year. If an existing request is found, the new request is blocked and an error message is displayed.

16.1 Creating the Business Rule

Navigation

All → System Definition → Business Rules

Steps

Navigate to All → System Definition → Business Rules.

Click New.

Enter the Business Rule name:

Laptop Request BR

Select the table:

Laptop Request Form

Enable the Advanced option.

Under When to Run, select:

When: Before

Insert: Enabled

Navigate to the Advanced section.

Add the Business Rule script.

Save the Business Rule.

16.2 Business Logic

When a new laptop request is submitted:

The system identifies the beginning and end of the current calendar year.

It searches the Laptop Request Form table.

It checks whether the same requester already has a request.

It checks whether the existing request was created during the current year.

If an existing request is found:

An error message is displayed.

The new record is prevented from being inserted.

If no existing request is found, the request is allowed.

16.3 Business Rule Script

```
(function executeRule(current, previous) {  
  
    var startOfYear = gs.beginningOfThisYear();  
    var endOfYear = gs.endOfThisYear();  
  
    // Check for existing request by same requester in the same year  
    var gr = new GlideRecord('u_laptop_request_form');  
  
    gr.addQuery('u_requester', current.u_requester);  
    gr.addQuery('sys_created_on', '>=', startOfYear);  
    gr.addQuery('sys_created_on', '<=', endOfYear);
```

```
gr.query();

if (gr.next()) {
    gs.addErrorMessage(
        'You have already submitted a laptop request for this year.'
    );

    current.setAbortAction(true);
}

})(current, previous);
```

Result

An employee who already has a laptop request in the current calendar year cannot create another request.

17. Role-Based Access Control

Purpose

Role-based access control was implemented to restrict application functionality according to the user's responsibilities.

Two roles were created:

it_manager

it_employee

17.1 Creating Roles

Navigation

All → User Administration / Users and Groups → Roles

Steps

Navigate to the Roles module.

Click New.

Create the required roles.

Save each role.

Roles Created

Role Name	Purpose
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it_manager	Provides view-only access to managers
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it_employee	Allows employees to create and manage laptop requests
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18. Creating User Groups

Purpose

Groups are used to manage users and assign appropriate roles efficiently.

Instead of assigning roles individually to every user, roles can be assigned to groups. Users who are members of those groups automatically receive the assigned roles.

18.1 Creating Groups

Navigation

All → User Administration / Users and Groups → Groups

Steps

Navigate to Groups.

Click New.

Create the required groups.

Add users as group members.

Assign the corresponding roles to each group.

Save the groups.

18.2 Groups Created

Group Name	Role Assigned	Members
IT Managers	it_manager	8 users
IT Employees	it_employee	10 users

Result

Members of IT Managers receive the it_manager role.

Members of IT Employees receive the it_employee role.

This simplifies role management and makes access administration easier.

19. Access Control Lists (ACLs)

Purpose

ACLs were implemented to restrict access to the Laptop Request Form table based on user roles.

The security design provides different permissions to managers and employees.

20. Manager Access – Read Only

Managers require access to view laptop requests but should not be able to create or modify them.

ACL Configuration

Property	Value
Table	Laptop Request Form
ACL Type	Table
Operation	Read
Role	it_manager
ACL Name	u_laptop_request_form.none

Permissions

The manager is allowed to:

View the Laptop Request Form records.

View the request list.

View request details.

The manager is not allowed to:

Create requests.

Modify requests.

Result

Managers have read-only access to laptop request records.

21. Employee Access – Full Request Management

Employees require permissions to create and manage laptop requests.

Three ACLs were created for the employee role.

ACL Configuration

Operation	Table	Role
Read	Laptop Request Form	it_employee
Create	Laptop Request Form	it_employee
Write	Laptop Request Form	it_employee

Employee Permissions

Employees can:

Create laptop requests.

View laptop request records.

Modify permitted request records.

View request status.

22. Creating ACLs

Navigation

All → System Security → Access Control (ACL)

Important Requirement

Before creating or modifying ACLs, the administrator needs the appropriate elevated security privileges.

Steps

Open the user profile.

Enable the required elevated role.

Navigate to:

All → System Security → Access Control (ACL)

Click New.

Select the required operation.

Select the Laptop Request Form table.

Specify the appropriate role.

Configure the ACL.

Save the ACL.

Repeat the process for the remaining operations.

23. ACL Design

Table-level ACLs were used with the following structure:

u_laptop_request_form.none

The .none ACL represents access to the table as a whole.

Manager ACL

u_laptop_request_form.none

|
└── Read
|

└─ it_manager

Employee ACLs

u_laptop_request_form.none

|

└─ Read → it_employee

└─ Create → it_employee

└─ Write → it_employee

Purpose

Table-level ACLs provide consistent access control across the records in the Laptop Request Form table.

24. Security Model

The application follows a role-based security model.

Manager

IT Manager

↓

it_manager role

↓

Read ACL

↓

Laptop Request Form

↓

View Only

Employee

IT Employee

↓

it_employee role



Read + Create + Write ACLs



Laptop Request Form



Create / View / Modify

25. Test Scenarios

The application was tested using different user roles and request scenarios.

Test Scenario	Expected Result
Employee creates a laptop request	Allowed
Employee views request	Allowed
Employee edits request	Allowed
Employee submits second request in same year	Blocked
Manager views request list	Allowed
Manager views request details	Allowed
Manager creates request	Blocked
Manager modifies request	Blocked

26. Business Rule Test

Scenario

An employee submits a laptop request for the first time during the current year.

Expected Result:

Request is successfully created.

Second Request

The same employee attempts to submit another laptop request during the same calendar year.

Expected Result:

You have already submitted a laptop request for this year.

The new record is not created.

27. Access Control Test

Employee Test

A user with the it_employee role attempts to create a laptop request.

Result: Allowed.

The employee can also view and modify the permitted request data according to the configured ACLs.

Manager Test

A user with the it_manager role opens the Laptop Request Form.

Result: The manager can view the records.

When the manager attempts to create or modify a record:

Result: Access is denied because no Create or Write permission has been provided to the manager role.

28. Security Components Summary

Component	Configuration	Purpose
Business Rule	Laptop Request BR	Prevent duplicate yearly requests
Role	it_manager	Manager access
Role	it_employee	Employee access
Group	IT Managers	Manage manager users
Group	IT Employees	Manage employee users
Read ACL	it_manager	Manager read access
Read ACL	it_employee	Employee read access
Create ACL	it_employee	Employee request creation
Write ACL	it_employee	Employee modification

29. Overall Application Flow

Employee



Employee Service Catalog



Laptop Request Record Producer



Laptop Request Form



Business Rule



Check Existing Request



Request exists?



Yes



No



Block Request



Create Request



Error Message



ACL Validation



Employee / Manager



Approval Process

30. Final Outcome

The Laptop Request Management application implements both business validation and role-based security.

The Business Rule ensures that an employee cannot submit more than one laptop request during the same calendar year. Role-based access control separates manager and employee permissions, while ACLs enforce these permissions at the table level.

The completed solution provides:

One laptop request per employee per year.

Backend validation using a Business Rule.

Separate manager and employee roles.

Group-based role assignment.

Manager read-only access.

Employee create, read, and write access.

Table-level ACL security.

Controlled access to laptop request records.

Integration with the Service Catalog and approval process.